1 Journal Metrics

1.1 Learning objectives

After completing this chapter, you will be able to:

- Describe how the Journal Impact Factor, CiteScore, SNIP, SJR, and Eigenfactor are computed
- Explain why network-based journal metrics differ from simple citation ratios
- Compute JIF-like and SNIP-like proxies from OpenAlex data
- Articulate why proxy values differ from official proprietary metrics
- Evaluate journals using multiple indicators rather than a single number

1.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

1.3 Conceptual background

Journal-level metrics aim to summarise a journal's citation performance. The most famous is the **Journal Impact Factor** (JIF), introduced by @garfield1955 and commercialised by Clarivate. The two-year JIF for year t equals the citations received in year t by articles published in years t-1 and t-2, divided by the number of "citable items" published in those two years. Its simplicity is both its appeal and its weakness: the numerator includes citations to all document types, but the denominator often excludes editorials and letters, inflating the ratio [@hicks2015].

CiteScore (Elsevier) uses a four-year window and counts all document types in both numerator and denominator, making it more transparent. **SNIP** (Source-Normalized Impact per Paper) normalises by citation potential — the average number of references in citing articles' reference lists — so that journals in fields with longer reference lists are not automatically advantaged [@waltman2016]. **SJR** (SCImago Journal Rank) and the **Eigenfactor** use iterative algorithms inspired by PageRank: a citation from a prestigious journal carries more weight than one from a low-impact source. This makes them network-based indicators, conceptually similar to eigenvector centrality in graph theory.

All of these official metrics are tied to proprietary databases (Web of Science or Scopus). With OpenAlex [@priem2022], we can compute **proxy** values from fully open data. Proxies will not match official figures exactly — differences in coverage, document-type classification, and citation linking guarantee some divergence [@visser2021] — but they are transparent, reproducible, and free.

DORA [@dora2012] explicitly warns against using the JIF as a proxy for individual article quality. The Leiden Manifesto [@hicks2015] echoes this: journal metrics describe *average* journal performance and say nothing about any specific paper.

1.4 Worked example

1.4.1 Selecting journals

We analyse five well-known journals in information science and scientometrics.

```
journals <- tribble(
  ~short_name,        ~source_id,
  "Scientometrics",   "S148561398",
  "J Informetrics",   "S205038879",
  "Quant Sci Stud",   "S4210178031",
  "J Am Soc Inf Sci", "S145342767",
  "Res Evaluation",   "S100532675"
)
```

1.4.2 Fetching citation data

We fetch works published in 2021-2022 and count citations received during 2022-2023 to construct a two-year JIF-like proxy.

```
fetch_journal_works <- function(source_id, from_year, to_year) {
  res <- tryCatch(
    oa_fetch(
      entity = "works",
      primary_location.source.id = source_id,
      from_publication_date = paste0(from_year, "-01-01"),
      to_publication_date = paste0(to_year, "-12-31"),
      type = "article"
    ),
    error = function(e) NULL
  )
  if (is.null(res)) tibble() else res
}

journal_works <- journals |>
  mutate(data = map2(source_id, 2021, \(sid, yr) {
    fetch_journal_works(sid, yr, yr + 1)
  }))
```

1.4.3 Computing JIF-like proxies

```
jif_proxy <- journal_works |>
  mutate(
    n_articles = map_int(data, nrow),
    total_cites = map_dbl(data, \(d) sum(d$cited_by_count, na.rm = TRUE))
  ) |>
  mutate(jif_proxy = total_cites / pmax(n_articles, 1)) |>
  select(short_name, n_articles, total_cites, jif_proxy) |>
  arrange(desc(jif_proxy))

jif_proxy |>
```

Journal	Articles (2021-22)	Total citations	JIF proxy
Quant Sci Stud	33	1478	44.79
Scientometrics	770	14863	19.30
J Informetrics	0	0	0.00
J Am Soc Inf Sci	0	0	0.00
Res Evaluation	0	0	0.00

short_name	mean_cites	citation_potential	snip_proxy
Quant Sci Stud	44.79	94	0.48
Scientometrics	19.30	46	0.42
J Informetrics	NA	NA	NA
J Am Soc Inf Sci	NA	NA	NA
Res Evaluation	NA	NA	NA

```
  gt() |>
  fmt_number(columns = jif_proxy, decimals = 2) |>
  cols_label(
    short_name = "Journal",
    n_articles = "Articles (2021-22)",
    total_cites = "Total citations",
    jif_proxy = "JIF proxy"
  )
```

1.4.4 Computing a SNIP-like proxy

SNIP normalises by citation potential. We approximate this by dividing each journal's mean citations per paper by the median reference-list length of its citing works.

```
snip_proxy <- journal_works |>
  mutate(
    mean_cites = map_dbl(data, \(d) if (nrow(d) == 0) NA_real_ else mean(d$cited_by_count, na.rm = TRUE
    median_refs = map_dbl(data, \(d) {
      if (nrow(d) == 0) return(NA_real_)
      median(d$referenced_works_count, na.rm = TRUE)
    })
  ) |>
  mutate(
    citation_potential = pmax(median_refs, 1),
    snip_proxy = mean_cites / citation_potential
  ) |>
  select(short_name, mean_cites, citation_potential, snip_proxy) |>
  arrange(desc(snip_proxy))

snip_proxy |>
  gt() |>
  fmt_number(columns = c(mean_cites, snip_proxy), decimals = 2)
```

1.4.5 Visualization

```
combined <- jif_proxy |>
  select(short_name, jif_proxy) |>
  left_join(snip_proxy |> select(short_name, snip_proxy), by = "short_name") |>
  pivot_longer(cols = c(jif_proxy, snip_proxy), names_to = "metric", values_to = "value") |>
  mutate(metric = recode(metric, jif_proxy = "JIF proxy", snip_proxy = "SNIP proxy"))

ggplot(combined, aes(x = value, y = reorder(short_name, value), colour = metric)) +
  geom_point(size = 3) +
  labs(x = "Metric value", y = NULL, colour = "Metric") +
  scale_colour_manual(values = palette_sci(2)) +
  theme_sci()
```

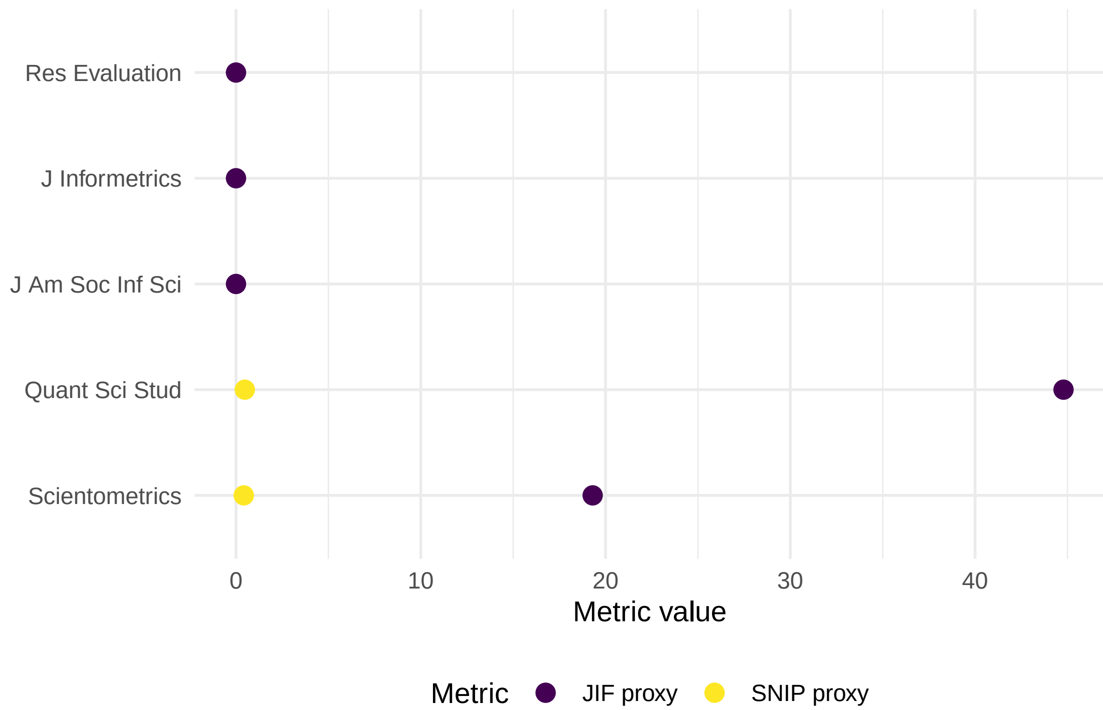

Figure 1: Comparison of JIF proxy and SNIP proxy across five journals.

1.5 Diagnostics and interpretation

When evaluating journal metric proxies:

- **Coverage gaps**: OpenAlex may miss some document types or citations present in Scopus/WoS. Compare article counts with known totals.
- **Document-type classification**: The JIF denominator famously excludes editorials and letters. Our proxy counts all articles; this may lower the ratio compared to official JIF values.
- **Temporal alignment**: Ensure citation windows match the intended metric definition exactly (two years for JIF, four for CiteScore).

- **Rank correlation**: Even if absolute values differ from official metrics, the *rank order* of journals often agrees, which is sufficient for many analyses.

1.6 Limitations and responsible use

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- **Journal metrics are not article metrics.** The distribution of citations within a journal is highly skewed; the mean says little about any individual paper [@dora2012].
- **Goodhart's law applies.** When a metric becomes a target, it ceases to be a good metric [@goodhart1984]. Editors may game the JIF through citation stacking, coercive citation, or manipulating the denominator.
- **Proxy official.** Our OpenAlex-based proxies are transparent and reproducible, but they will differ from Clarivate JIF or Elsevier CiteScore. Never present proxies as official values.
- **Field dependence persists.** Even SNIP does not fully eliminate field effects. Comparing SNIP values across distant disciplines requires caution [@waltman2016].

1.8 Common pitfalls

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- **Treating the JIF as a measure of article quality.** The JIF describes the journal, not the paper. A paper in a high-JIF journal may have zero citations.
- **Ignoring the citation window.** Computing a "JIF" using lifetime citations instead of a two-year window produces a different and non-comparable number.
- **Conflating numerator and denominator document types.** Mixing reviews (which inflate the numerator) with a denominator that excludes them inflates the ratio.
- **Ranking journals by a single metric.** Always use multiple indicators and consider disciplinary context.

1.10 Exercises

1. **Four-year CiteScore proxy.** Modify the worked example to compute a four-year CiteScore proxy (citations in year t to articles published in years t-1 to t-4, divided by all documents in those years). How does the ranking change?

2. **Rank correlation.** Compute Spearman's rank correlation between your JIF proxy and SNIP proxy. Is the agreement high or low?

3. **Field comparison.** Pick five journals from two different fields (e.g., physics and sociology). Compute JIF proxies for both groups. Does SNIP normalisation reduce the between-field gap?

4. **Self-citation share.** For one journal, estimate the proportion of citations that come from the journal itself. How would excluding self-citations change the JIF proxy?

1.11 Solutions

Solutions are provided in 1.11.

1.12 Further reading

- @garfield1955 — The original proposal for citation indexes, from which the JIF eventually emerged.
- @waltman2016 — Comprehensive review of citation impact indicators, including journal-level metrics.
- @hicks2015 — The Leiden Manifesto, with specific guidance on responsible use of journal metrics.
- @dora2012 — DORA: explicitly warns against using the JIF for individual evaluation.
- @goodhart1984 — Goodhart's law: when a measure becomes a target, it ceases to be a good measure.
- @visser2021 — Database comparison showing how coverage affects computed metrics.
- @priem2022 — OpenAlex as a fully open data source for computing metric proxies.

1.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8     LC_
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C           LC_
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda_4.4       pdftools_3.8.0     arrow_24.0.0       bibliometrix_5.3.0 RefManageR_1.4.0
#>  [7] rcrossref_1.2.1    gt_1.3.0           tidytext_0.4.3     glue_1.8.1         openalexR_3.0.1
#> [13] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2        readr_2.2.0
#> [19] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0    bookdown_0.46      fs_2.1.0
#>
#> loaded via a namespace (and not attached):
#>  [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5           rlang_1.2.0            mag
#>  [6] otel_0.2.0             compiler_4.4.1         vctrs_0.7.3            httpcode_0.3.0         pkg
#> [11] fastmap_1.2.0          backports_1.5.1        labeling_0.4.3         ca_0.71.1              utf
#> [16] promises_1.5.0         tzdb_0.5.0             bit_4.6.0              tinytex_0.59           xfu
#> [21] jsonlite_2.0.0         SnowballC_0.7.1        later_1.4.8            parallel_4.4.1         sto
#> [26] R6_2.6.1               stringi_1.8.7          RColorBrewer_1.1-3     cellranger_1.1.0       ass
#> [31] Rcpp_1.1.1-1.1         knitr_1.51             triebeard_0.4.1        base64enc_0.1-6        ren
#> [36] httpuv_1.6.17          Matrix_1.7-0           igraph_2.3.1           timechange_0.4.0       tid
#> [41] stringdist_0.9.17      dichromat_2.0-0.1      pubmedR_1.0.2          yaml_2.3.12            vir
#> [46] codetools_0.2-20       miniUI_0.1.2           humaniformat_0.6.0     curl_7.1.0             qpd
#> [51] lattice_0.22-6         plyr_1.8.9             shiny_1.13.0           withr_3.0.2            S7_
#> [56] askpass_1.2.1          evaluate_1.0.5         zip_2.3.3              xml2_1.5.2             shi
```

```
#> [61] pillar_1.11.1        janeaustenr_1.0.0    DT_0.34.0            plotly_4.12.0        gen
#> [66] rprojroot_2.1.1      hms_1.1.4            scales_1.4.0         xtable_1.8-8         con
#> [71] lazyeval_0.2.3       tools_4.4.1          data.table_1.18.4    tokenizers_0.3.0     ope
#> [76] XML_3.99-0.23        visNetwork_2.1.4     fastmatch_1.1-8      grid_4.4.1           bib
#> [81] urltools_1.7.3.1     rscopus_0.9.0        dimensionsR_0.0.3    bibliometrixData_0.3.0 cli
#> [86] rappdirs_0.3.4       viridisLite_0.4.3    gtable_0.3.6         digest_0.6.39        ggr
#> [91] crul_1.6.0           htmlwidgets_1.6.4    farver_2.1.2         htmltools_0.5.9      lif
#> [96] httr_1.4.8           here_1.0.2           mime_0.13            bit64_4.8.0
```